

Aarya Pant

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PROFILE

Geomatics Engineer with a strong foundation in GIS, remote sensing, and web development. Interested in combining GIS, UAV, Web based technologies to solve environmental issues and infra planning. Seeking opportunities related to GIS, Remote Sensing, Geospatial Data Analysis, Mapping and Surveying. Eager to apply technical expertise in spatial analysis and data visualization to solve real-world challenges.

SKILLS

- GIS
- Remote Sensing
- Photogrammetry
- Surveying
- Web GIS
- Web Development
- Hec-HMS
- Hec-RAS
- Python

EDUCATION

Kathmandu University | *BE in Geomatics Engineering*

Dhulikhel | **Nov 2021 – Feb 2026**

- Semester-wise GPA:

1st Sem: 2.88

2nd Sem: 3.36

3rd Sem: 3.85

4th Sem: 3.88

5th Sem: 4.00

6th Sem: 3.85

7th Sem: 3.96

8th Sem: 4

- Cumulative GPA: 3.71

NASA Higher Secondary School | *High School (NEB +2)*

Tinkune | **April 2019 – May 2021**

- Cumulative GPA: 3.46

Jaycees Everest English School | *Schooling (SEE)*

Dhangadhi | **2009 – 2019**

- Cumulative GPA: 3.45

EXPERIENCE

GIS Expert | **Ucchakoti Nirman Sewa**

January 2026 – Present

- Worked under the Urban Water Supply and Sanitation Project (UWSSP), a project of the Department of Water Supply and Sewage Management, Panipokhari, Kathmandu, contributing to GIS data preparation, spatial analysis, and mapping.
- Collaborated with project stakeholders including M/S Kalika-Uchchakoti J/V and ITECO-UNEC J/V to support geospatial components of the project.

Intern | **DroNepal**

Nov 2025 – January 2026

- Conducted GIS mapping and analysis (LULC, topographical, contour, socio-infrastructure) for hydropower and road projects, including digitization and cadastral verification tasks.
- Processed UAV imagery and LiDAR data to generate orthophotos and spatial datasets using tools like LiGeoReference and LiDAR360.

- Applied advanced GIS techniques and enhanced the company's Web GIS platform using HTML, CSS, and JavaScript, along with supporting technical documentation and educational content preparation.

Student Participant | Australia-Nepal Mobility Program

Aug 15 – Aug 25

- Participated in a month-long academic and cultural exchange program with the University of Southern Queensland (UniSQ) under the New Colombo Plan (NCP). Conducted joint fieldwork with UniSQ and LMTC, including instrument calibration, levelling, topographic and cadastral surveys, data processing, and drone applications.

Treasurer | Geomatics Engineering Society

Sept 2024 – Dec 2025

- Managed society finances with transparency, prepared budgets, and supported event organization and coordination.

Trainer | Geowomen: Promoting Spatial Data Literacy

Feb 2025

- Led a two-day virtual training session organized by NAXA, Robotics Association of Nepal (RAN), and Women in Geospatial Nepal (WiG Nepal).
- Facilitated hands-on mapping activities using OpenStreetMap (OSM) and the OSM Tasking Manager.
- Guided participants in mapping buildings and roads to enhance rural and urban accessibility.

G17 Global | G17 University Ambassador for SDG 5 (Gender Equality)

June 2024 – June 2025

- Advocated for gender equality and women's empowerment through awareness, campaigns, and student engagement initiatives at the university level.

Sudurpaschim Samaj (SuPa Samaj, KU) | Media and Marketing Head

Feb 2024 – Dec 2025

- Led media, publicity, and outreach initiatives, managed social platforms, and promoted cultural and academic events to strengthen community engagement.

PROJECTS

- **Modelling and Mapping Flood Induntation and Risk using a 2D Hydraulic Approach** [docs](#)
This study focuses on simulating and mapping flood inundation and risk in the Barabise area using a two-dimensional hydraulic modelling approach to assess flood dynamics and support disaster risk management.
- **Vulnerability and Risk Zonation Dashboard** [docs](#)
Developed a mini geoportal integrating hazard, exposure, and vulnerability datasets for interactive flood and disaster risk visualization using open-source web technologies.
- **Accuracy Assessment and Comparison of DEM Interpolation Techniques in GIS** [docs](#)
Conducted spatial accuracy analysis of different DEM interpolation methods and evaluated the impact of terrain characteristics on model accuracy.
- **Flood Risk Assessment of Manahari Khola, Nepal Using HEC-RAS Hydraulic Modelling**
Performed hydrodynamic modelling for flood hazard mapping and integrated HEC-RAS and GIS for flood extent and risk zone analysis.
- **AQI Monitoring Dashboard** [docs](#)
Designed a real-time air quality monitoring dashboard with interactive charts to visualize pollutant levels, AQI trends, and weather parameters for data-driven environmental insights.
- **Windfarm Site Selection Suitability using AHP: Myagdi District** [docs](#)
Conducted multi-criteria site selection using AHP, integrating wind, slope, land use, and accessibility data to identify optimal wind energy sites.

- **Shortest Route Analysis using PgRouting: Kathmandu to Dhangadhi** [docs](#)
Performed network analysis using PgRouting to determine the most efficient routes based on road networks and distances.
- **Konvertor - Coordinate Systems Tool** [docs](#)
Built a tool for coordinate conversion and system transformation to improve geospatial data interoperability and accuracy in JAVA.
- **Impact of Land Fragmentation on Agriculture** [docs](#)
Studied how land fragmentation affects agricultural productivity in Nepal, analysing field and secondary data to suggest sustainable management strategies.
- **NDVI Analysis of Dhulikhel** [docs](#)
Applied remote sensing to assess vegetation health and land cover changes, generating NDVI maps for monitoring ecological trends.
- **Open Mapping Hub Asia-Pacific Field Participant: Urban Rooftop Farming**
Conducted fieldwork in Banepa Municipality, collecting geospatial data and supporting sustainability assessments for urban rooftop farming.
- **Survey Camp Projects:** Topographic Surveying, Levelling (Differential, Profile and Cross Section), Road and Bridge Site Surveying, Road Alignment Planning, Plane Table Survey, Drone Survey, GNSS Survey, Cadastral Survey, Triangulation Survey, Hydrographical Survey

TECHNOLOGIES

- Languages: C++, C, SQL, JavaScript, Python, Java
- Technologies: QGIS, Envi, PiX4D, Hec-RAS, Hec-HMS, Microsoft Office Suite, Flask, Torch, NumPy, Parcel Editor

SOFT SKILLS

- Communication: Public speaking, Negotiation, Active listening.
- Leadership Teamwork: Decision-making, Conflict resolution, Collaboration.
- Problem-Solving: Analytical thinking, Creativity, Troubleshooting.
- Time Management: Prioritization, Multitasking, Meeting deadlines.
- Adaptability: Handling pressure, Flexibility, Crisis management